BizQuery Project Stage 2

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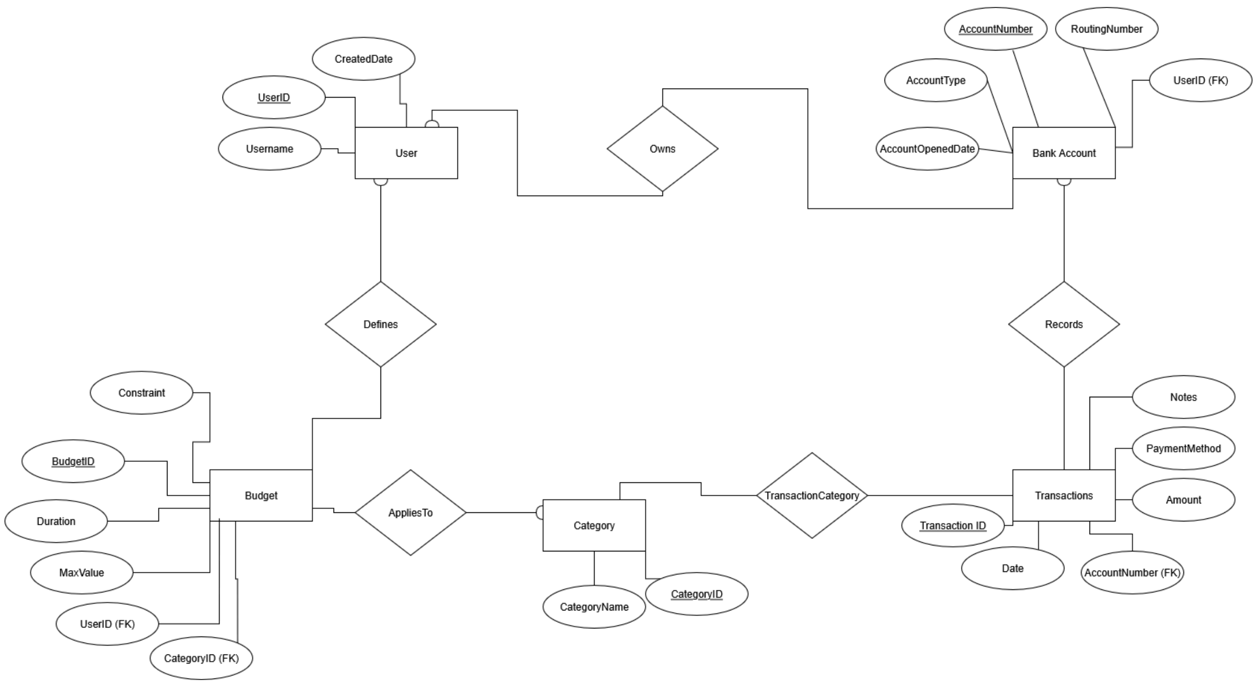
ER Diagram



Assumptions

User

The user was assigned as an entity instead of an attribute because a user manages the accounts, budgets, and categories. It is central to the entire database and project to have this entity.
Username was chosen to be a VARCHAR to keep track of a UserID's username. The CreatedDate is used to track when the user's profile was created.

BankAccount

The bank accounts of the users were modeled as an entity because accounts have distinct attributes and relationships to transactions. These attributes include account number, routing number, account type, and date opened.
The AccountNumber is the primary key because it is unique per bank account. The VARCHAR domain will allow for storage of alphanumeric numbers associated with the account.

Category
The category for budgets and transactions is modeled as an entity because both budgets and transactions reference similar categories. It makes sense to have this be a shared entity between the two entities.
The CategoryID is the primary key because it is unique for each individual category. The INT domain will allow for easy reference within this entity and others.

Transactions
Transactions are modeled as an entity because it records historical events tied to bank accounts, categories, and payment methods.
The TransactionID is the primary key to keep track of each transaction. Amount is stored as a DECIMAL(10,2) to handle the currency in the transaction.

Budget
Budgets are modeled as an entity because multiple budgets exist per user and they reference different categories as well.
The BudgetID is used as the primary key to keep track of unique budgets. The MaxValue is stored as a DECIMAL(10,2) to represent money. The Duration is stored as a VARCHAR(20) to represent periods of time ("Yearly", "Monthly").

Relationships & Cardinality

For the cardinality of the **Defines** relationship in the diagram, one user can own many accounts and define many budgets. Each account or budget belongs to exactly one user.
For the cardinality of the **Owns** relationship, a user may own many accounts, and each account belongs to exactly one user.
For the cardinality of the **TransactionCategory** relationship, a category may classify many transactions, and each transaction may fall under many categories.
For the cardinality of the **Records** relationship, one bank account can have many transactions, but one transaction belongs to exactly one bank account.
For the cardinality of the **AppliesTo** relationship, one category may have many budgets belonging to it, but one budget belongs to exactly one category.

User - BankAccount: 1-to-many
BankAccount - Transactions: 1-to-many
Transactions - Category: many-to-many
User - Budget: 1-to-many
Budget - Category: many-to-1

Normalization

A relation is in BCNF if for every functional dependency (FD), the determinant is a superkey.
We can check this for each entity:

For **User**, FD: UserID -> Username, CreatedDate. UserID is the primary key, so the determinant is a superkey.
For **BankAccount**, FD: AccountNumber->UserID, RoutingNumber, AccountType, AccountOpenedDate. AccountNumber is the primary key, so the determinant is a superkey.
For **Category**, FD: CategoryID -> CategoryName. CategoryID is the primary key, so the determinant is a superkey.
For **Transactions**, FD: TransactionID -> AccountNumber, Date, Amount, PaymentMethod, Notes. TransactionID is the primary key, so the determinant is a superkey.
For **TransactionCategory**, FD: (TransactionID, CategoryID) → nothing. The determinant is the full primary key, which is a candidate key.
For **Budget**, FD: BudgetID -> UserID, CategoryID, Duration, MaxValue, Constraint. BudgetID is the primary key, so the determinant is a superkey.
All relations have only functional dependencies where the determinant is a primary key or a candidate key. The entire schema is in BCNF form.

Logical Design - Relational Schema

User(UserID: INT [PK], Username: VARCHAR(50), CreatedDate: DATE)

BankAccount(AccountNumber: VARCHAR(20) [PK],
    UserID: INT [FK to User.UserID],
    RoutingNumber: VARCHAR(20),
    AccountType: VARCHAR(20),
    AccountOpenedDate: DATE)

Category(CategoryID: INT [PK], CategoryName: VARCHAR(50))

Transactions(TransactionID: INT [PK],
    AccountNumber: VARCHAR(20) [FK to BankAccount.AccountNumber],
    Date: DATE,
    Amount: DECIMAL(10,2),
    PaymentMethod: VARCHAR(30),
    Notes: VARCHAR(255))

TransactionCategory(TransactionID: INT [FK to Transaction.TransactionID],
    CategoryID: INT [FK to Category.CategoryID],
    [PK = (TransactionID, CategoryID)])

Budget(BudgetID: INT [PK],
    UserID: INT [FK to User.UserID],
    CategoryID: INT [FK to Category.CategoryID],
    Duration: VARCHAR(20),
    MaxValue: DECIMAL(10,2),
    Constraint: VARCHAR(50))